

Pseudocode:

1. start
2. Ask the user: "Vehicle type (E = Electric, H = Hybrid)"
3. Read input and store it in type
4. Ask the user: "Current battery level (%)"
5. Read input and store it in soc
6. Ask the user: "Required charging level (%)"
7. Read input and store it in required
8. Ask the user: "Expected parking duration (hours)"
9. Read input and store it in duration
10. Ask the user: "Current time (HH MM)"
11. Read input and store it in hour and minute
12. Ask the user: "Parking membership (Y/N)"
13. Read input and store it in member
14. Ask the user: "Disabled-person priority (Y/N)"
15. Read input and store it in disabled
16. Ask the user: "Charging station available (Y/N)"
17. Read input and store it in station
18. Set type, member, disabled, station to uppercase
19. IF any input is invalid THEN
 Display: "Invalid input. Please restart and enter valid values."
 End
ENDIF
20. IF duration <= 2 THEN

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    Set parkingBase = 200
ELSE IF duration <= 5 THEN
    Set parkingBase = 400
ELSE
    Set parkingBase = 700
ENDIF

21. Set parkingDiscount = 0
22. IF disabled = Y THEN
    Set parkingDiscount = parkingBase
    Set parkingCost = 0
ELSE IF member = Y THEN
    Set parkingDiscount = parkingBase * 0.20
    Set parkingCost = parkingBase - parkingDiscount
ELSE
    Set parkingCost = parkingBase
ENDIF

23. IF duration > 8 THEN
    Set message = "Long-stay warning: Please relocate your vehicle after
charging."
ELSE
    Set message = "Standard parking duration."
ENDIF

24. IF station = N THEN
    IF type = H THEN
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Display: "Charging unavailable - Parking only."

Display: parkingCost, parkingDiscount, final amount = parkingCost,
message

ELSE

Display: "No charging slot available."

ENDIF

End

ENDIF

25. IF type = H AND soc >= 40 THEN

Display: "Vehicle does not qualify for EV charging."

End

ENDIF

26. IF required <= soc THEN

Display: "No charging required."

End

ENDIF

27. Set requiredCharging = required - soc

28. Set emergency = FALSE

29. IF soc <= 15 AND required >= 80 THEN

Set priority = "Emergency Charging Priority"

Set emergency = TRUE

ELSE IF disabled = Y OR (member = Y AND soc <= 30) THEN

Set priority = "Priority Charging"

ELSE

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    Set priority = "Normal Charging"
ENDIF
30. Set timeInMin = hour * 60 + minute
31. IF timeInMin >= 1020 AND timeInMin <= 1320 THEN
    Set status = "Peak"
    Set rate = 50
ELSE
    Set status = "Off-peak"
    Set rate = 35
ENDIF
32. Set chargingBase = requiredCharging * rate
33. Set chargingDiscount = 0
34. IF member = Y AND emergency = FALSE THEN
    IF status = "Peak" THEN
        Set chargingDiscount = chargingBase * 0.10
    ELSE
        Set chargingDiscount = chargingBase * 0.20
    ENDIF
ENDIF
35. Set chargingCost = chargingBase - chargingDiscount
36. Set totalDiscount = chargingDiscount + parkingDiscount
37. Set finalAmount = chargingCost + parkingCost
38. Display: type, soc, requiredCharging, priority, status
39. Display: chargingCost, parkingCost, totalDiscount, finalAmount
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40. Display: message

41. End

IPO Chart:

<u>INPUT</u>	<u>PROCESS</u>	<u>MODULE REFERENCE</u>	<u>OUTPUT</u>
Type	1.Enter vehicle type	Read	Vehicle type
Soc	2.Enter battery level	Read	Current battery %
Required	3.Enter required charging level	Read	Required charging %
Duration	4.Enter parking duration	Read	Charging priority
Hour	5.Enter current time(hour, minutes)	Read	Peak / off-peak status
Minute	6.Enter membership status	Read	Charging cost
Member	7.Enter disabled priority rights	Read	Parking cost
Disabled	8.Enter station availability	Read	Total discount
station	9.Validate all inputs	Validate	Final payable amount
	10.Calculate parking cost and parking discount	Calc	Warning / message
	11.Set duration message	CheckDuration	

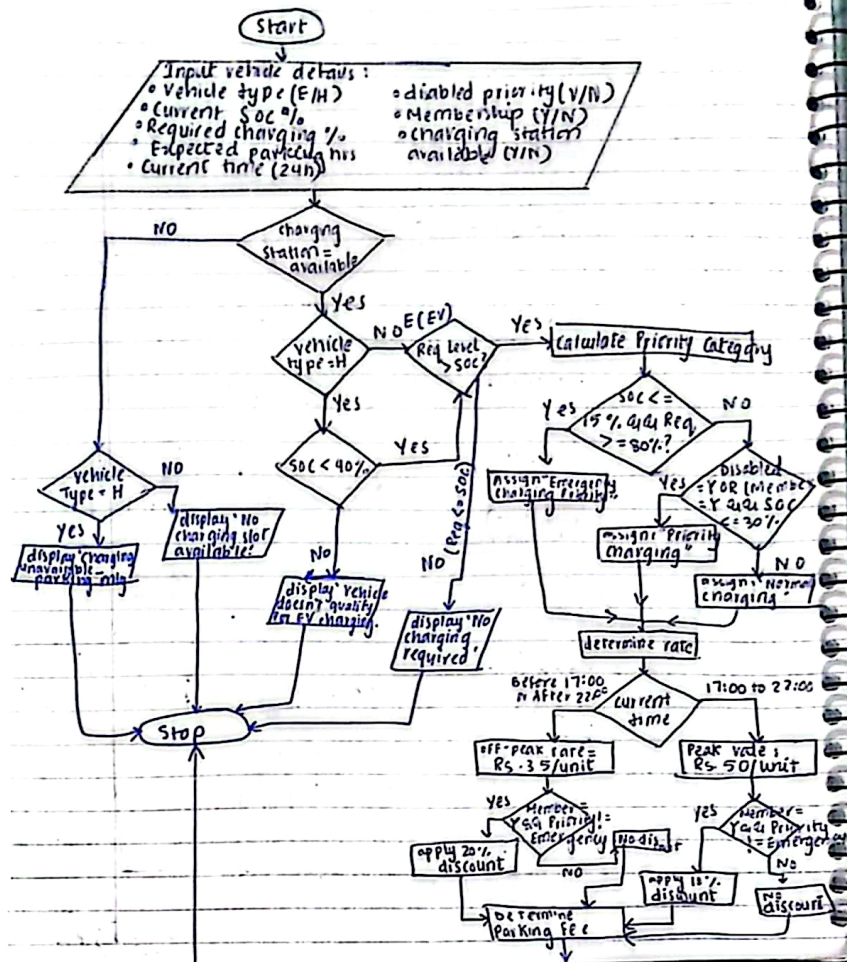
	12.Check station availability	CheckStation	
	13.Check vehicle eligibility	CheckEligibility	
	14.Calculate required charging	Calc	
	15.Assign charging priority	AssignPriority	
	16.Determine peak/off-peak	CheckTime	
	17.Calculate charging cost and discount	Calc	
	18.Calculate total discount and final amount	Calc	
	19.Print Results	Print	
	20.End	EVChargingControl	

PAC Table:

<u>Given Data</u>	<u>Required Result</u>	<u>Processing Required</u>	<u>Solution Alternative</u>
Vehicle type (E = Electric, H = Hybrid)	Vehicle type	Validate inputs	Use nested IF-ELSE statements for all decisions (used in this solution).
Battery level (SOC %)	Current battery percentage	If station unavailable: hybrid → "Charging unavailable Parking only.", electric → "No charging slot available."	Use a separate function for each calculation, such as parking cost and charging cost.
Required charging level (%)	Required charging percentage	If hybrid and battery $\geq 40\%$: "Vehicle does not qualify for EV charging."	Use a structure (struct) to store the customer data.
Expected parking duration (hours)	Charging priority	Required Charging = Required Level – Current Battery	Use a switch statement for vehicle type and priority.
Current time (24-hour, HH MM)	Peak / off-peak status	If Required Level $\leq$ Battery → "No charging required."	Use a lookup table for the rates and parking tariffs.

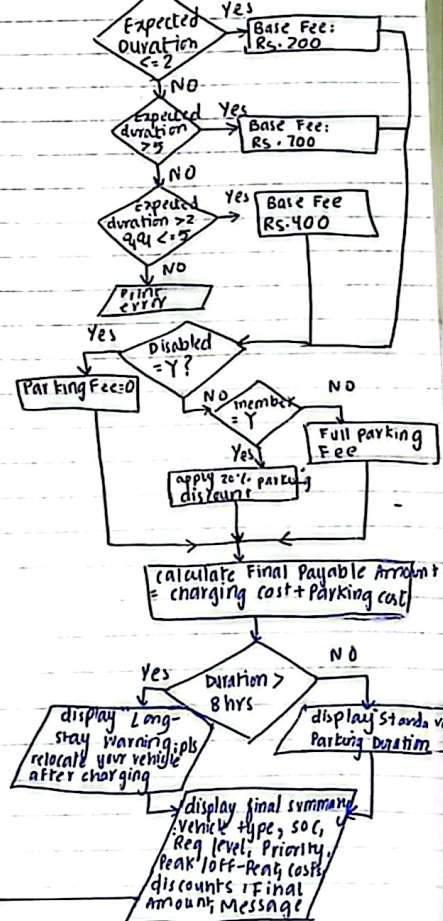
Parking membership (Y/N)	Charging cost	If Required Level $\leq$ Battery $\rightarrow$ "No charging required."	
Disabled-person priority (Y/N) • Charging	Parking cost	Peak if time is 17:00 to 22:00, otherwise off-peak 8. Charging Cost = Required Charging $\times$ Rate – Charging Discount	
Charging station available (Y/N)	Discount	Charging Discount = 20% (off-peak) or 10% (peak) for members; none for Emergency	
Fixed data in the problem: • Charging rate: Rs. 35 off-peak, Rs. 50 peak per unit • Parking: Rs. 200 (up to 2 h), Rs. 400 (up to 5 h), Rs. 700 (over 5 h)	Final payable amount	Parking Cost = Base Parking Charge – Parking Discount	

	Appropriate warning / message	Parking Discount = 100% for disabled, 20% for members	
		Total Discount = Charging Discount + Parking Discount	
		Final Amount = Charging Cost + Parking Cost	
		If duration > 8 hours : long-stay warning	



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Algorithm:

1. Ask the user to enter **Vehicle type** (E or H)
2. Ask the user to enter **Battery level**
3. Ask the user to enter **Required charging level**
4. Ask the user to enter **Parking duration**
5. Ask the user to enter **Current time** (hour and minute)
6. Ask the user to enter **Membership status** (Y/N)
7. Ask the user to enter **Disabled priority status** (Y/N)
8. Ask the user to enter **Station availability** (Y/N)
9. If any input is invalid, display "Invalid input" and stop
10. Set **Parking Base** to 200 if duration is 2 hours or less, to 400 if duration is 5 hours or less, otherwise to 700
11. Set **Parking Cost** to 0 and **Parking Discount** to the full Parking Base if the customer is disabled
12. Otherwise, if the customer is a member, set **Parking Discount** to 20% of Parking Base and **Parking Cost** to Parking Base – Parking Discount
13. Otherwise, set **Parking Cost** to Parking Base
14. Set the **Message** to the long-stay warning if duration is more than 8 hours, otherwise to "Standard parking duration"
15. If the station is unavailable:
 - If the vehicle is a hybrid, display "Charging unavailable – Parking only." with the parking cost, discount and final amount, then stop
 - Otherwise, display "No charging slot available." and stop
16. If the vehicle is a hybrid and battery is 40 or more, display "Vehicle does not qualify for EV charging." and stop
17. If Required level is less than or equal to Battery, display "No charging required." and stop
18. Set **Required Charging** to Required level – Battery
19. If battery is 15 or below and required level is 80 or above, set **Priority** to Emergency Charging Priority
20. Otherwise, if the customer is disabled, or is a member with battery of 30 or below, set **Priority** to Priority Charging
21. Otherwise, set **Priority** to Normal Charging
22. Set **Time** to hour × 60 + minute
23. If Time is between 17:00 and 22:00, set **Status** to Peak and **Rate** to 50, otherwise set **Status** to Off-peak and **Rate** to 35
24. Set **Charging Base** to Required Charging × Rate
25. If the customer is a member and Priority is not Emergency, set **Charging Discount** to 10% of Charging Base in peak hours or 20% in off-peak hours, otherwise set it to 0
26. Set **Charging Cost** to Charging Base – Charging Discount
27. Set **Total Discount** to Charging Discount + Parking Discount
28. Set **Final Amount** to Charging Cost + Parking Cost
29. Display vehicle type, battery, required charging, priority, peak or off-peak status, charging cost, parking cost, total discount, final amount and the message
30. Stop